Principles of Macroeconomics

Econ 2

Lecture 1:

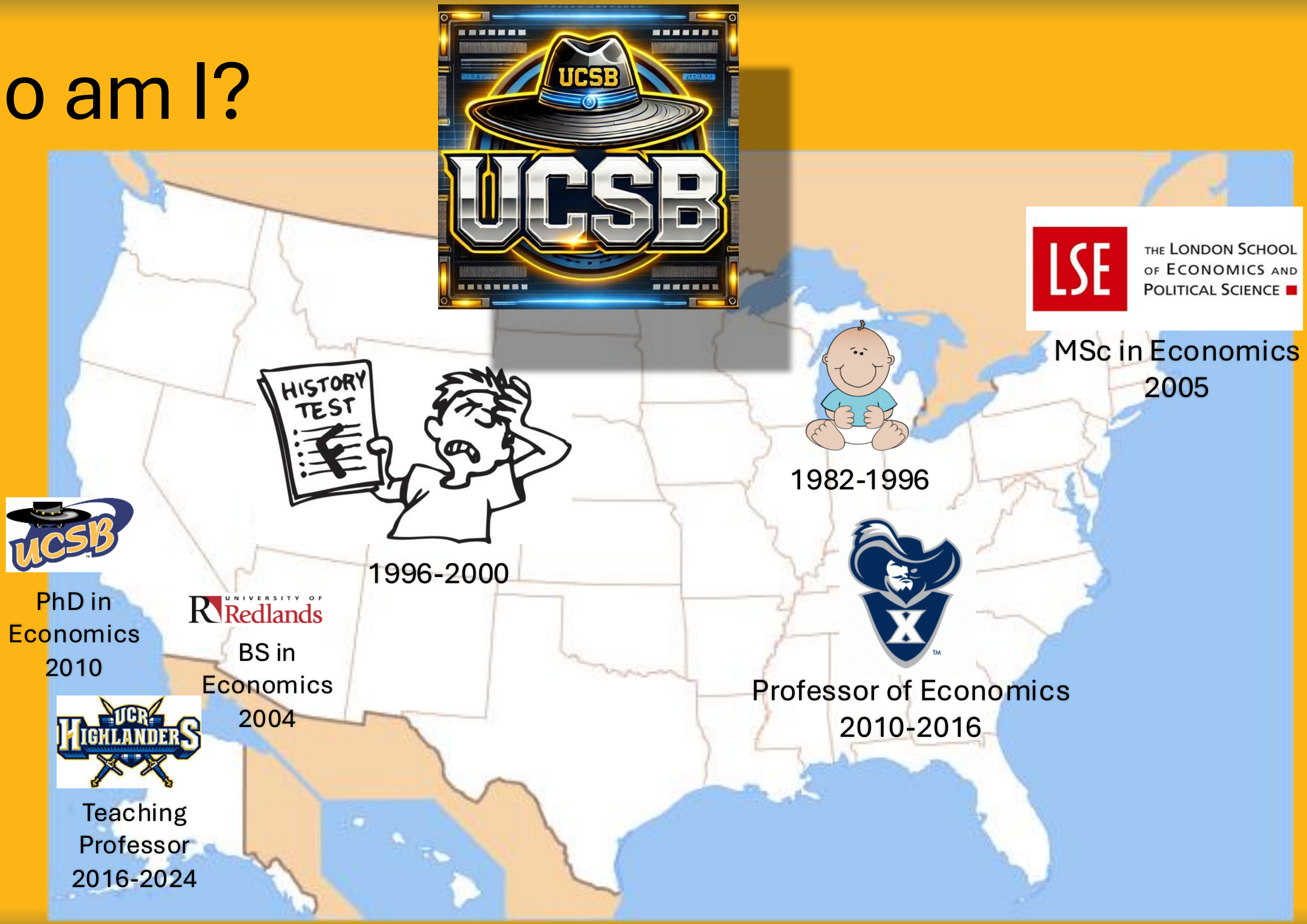
Course Introduction

Who am I?

Personal Website

- **Professor: Matt Lang**
- **Research Interests: Health Economics, Mental Health, Education Policy, Crime, Blockchain Technology**
- **Teaching Experience: Principles of Macro and Microeconomics, Stats, Econometrics, Game Theory, Health Economics, Macro, Macro, and more Macro!**
- **Unlikely Fact: Started playing ice hockey at 4 years old. Married another economist!**
- **Office Hours: North Hall 3032, Tuesday from 4 – 5:30pm**
- **“Coffee Hours: UCSB Campus/TBD**
  - **Random Sign-Up Times**
- **Email: langm@ucsb.edu**

Who am I?

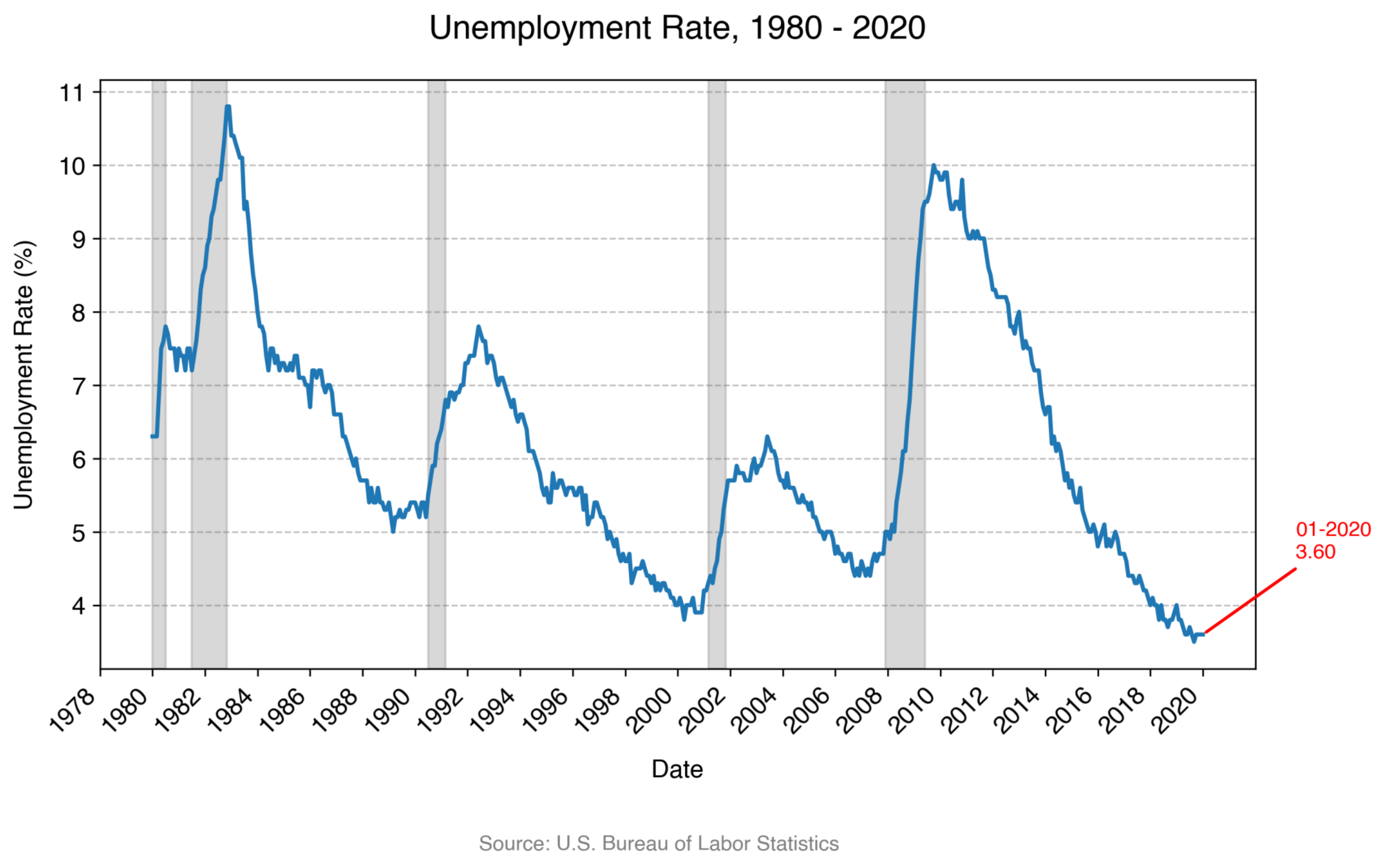

UCSB
UCSB
HISTORY TEST
1996-2000
1982-1996
LSE THE LONDON SCHOOL OF ECONOMICS AND POLITICAL SCIENCE
MSc in Economics
2005
UCSB
PhD in
Economics
2010
UNIVERSITY OF Redlands
BS in
Economics
2004
UCR HIGHLANDERS
Teaching
Professor
2016-2024
Professor of Economics
2010-2016

Course Overview

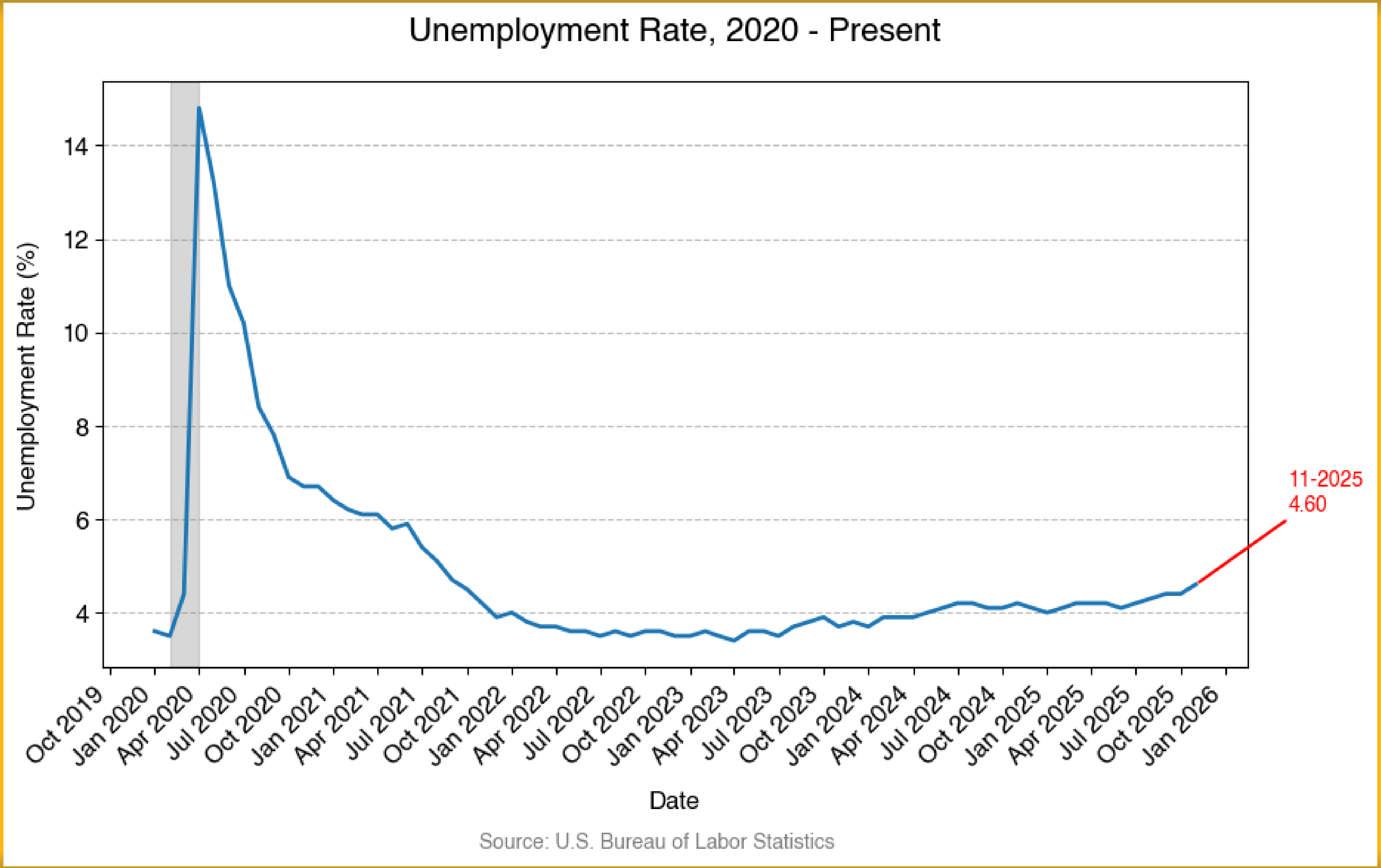
- Explore and understand fluctuations in the economy as a whole and assess economic policies in an **analytical** and **unbiased** way
  - Theory
  - Data
  - Conversation

- Course Guided by 3 Principles:
  - High Standard of Living
  - Stable Prices
  - Full Employment

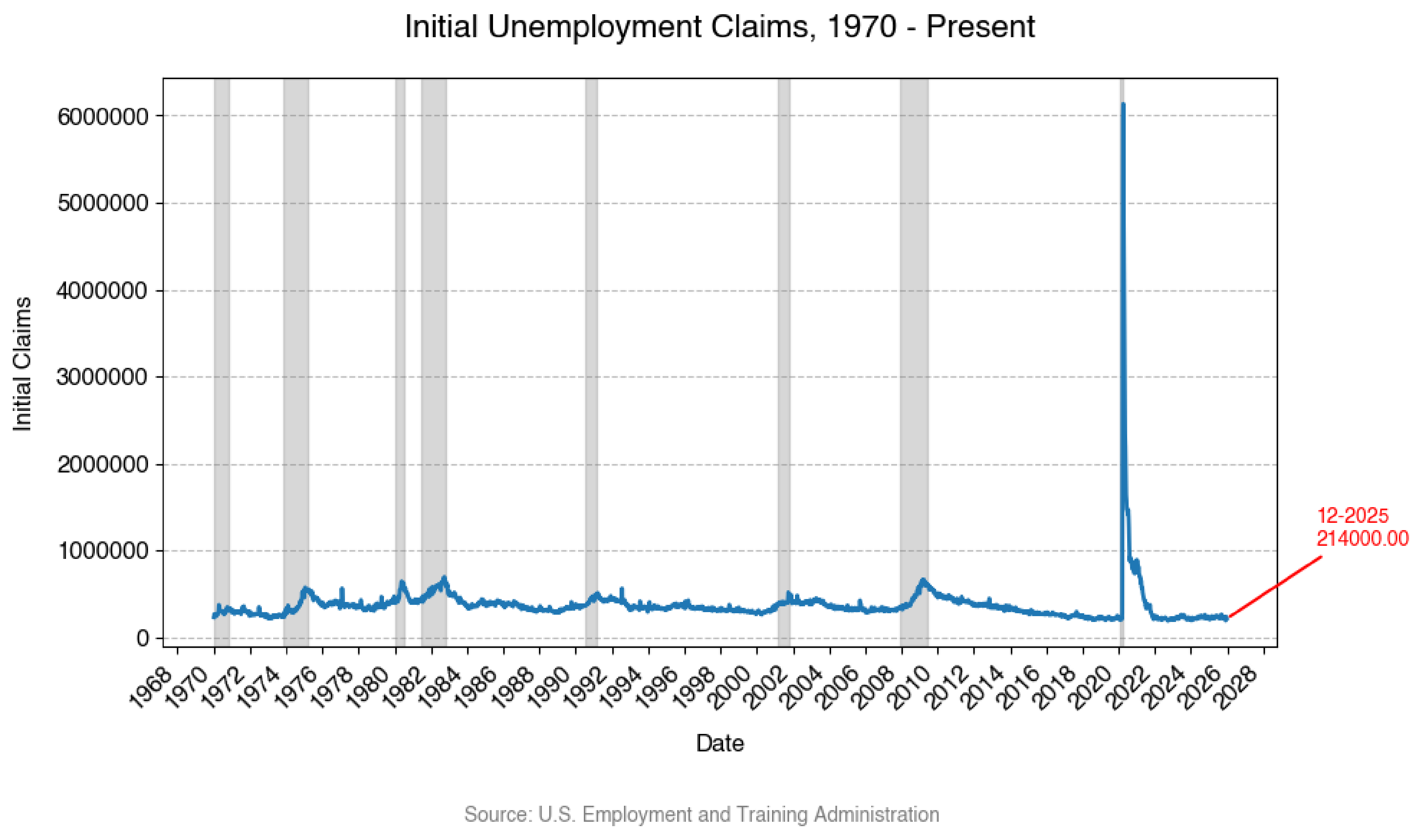
Course Website:
https://matthewdlang18.github.io/principles-of-macroeconomics/


Unemployment Rate, 1980 - 2020
Unemployment Rate (%)
11
10
9
8
7
6
5
4
1978
1980
1982
1984
1986
1988
1990
1992
1994
1996
1998
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
Date
01-2020
3.60
Source: U.S. Bureau of Labor Statistics


Unemployment Rate, 1990 - Present
Unemployment Rate (%)
14
12
10
8
6
4
1990
1992
1994
1996
1998
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
2026
Date
02-2025
4.10
Source: U.S. Bureau of Labor Statistics

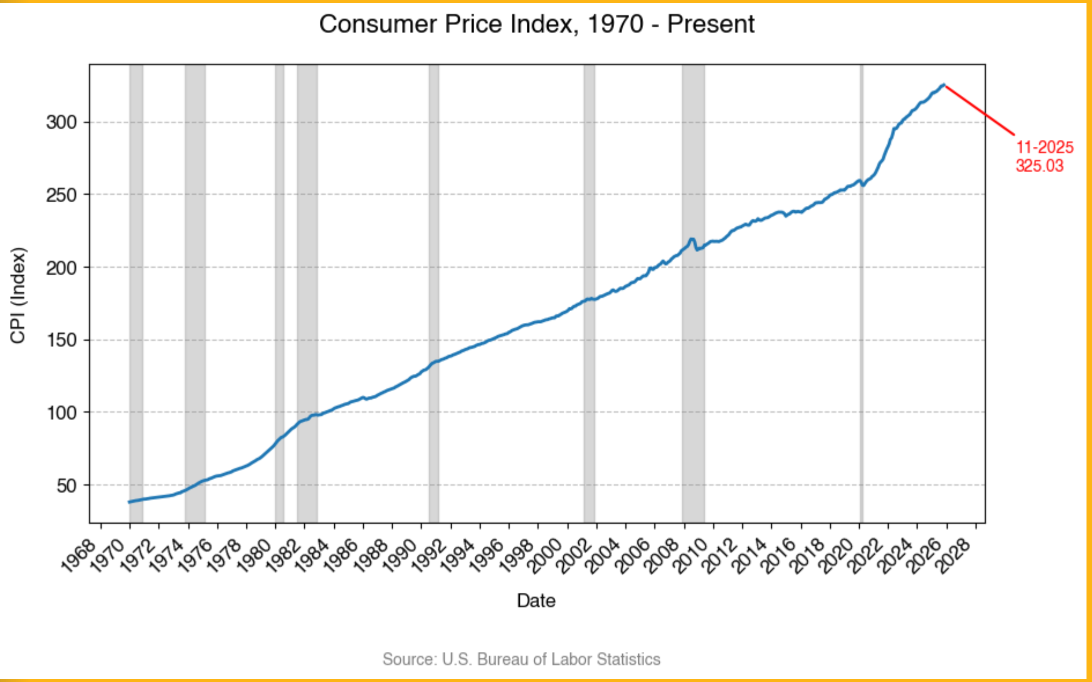


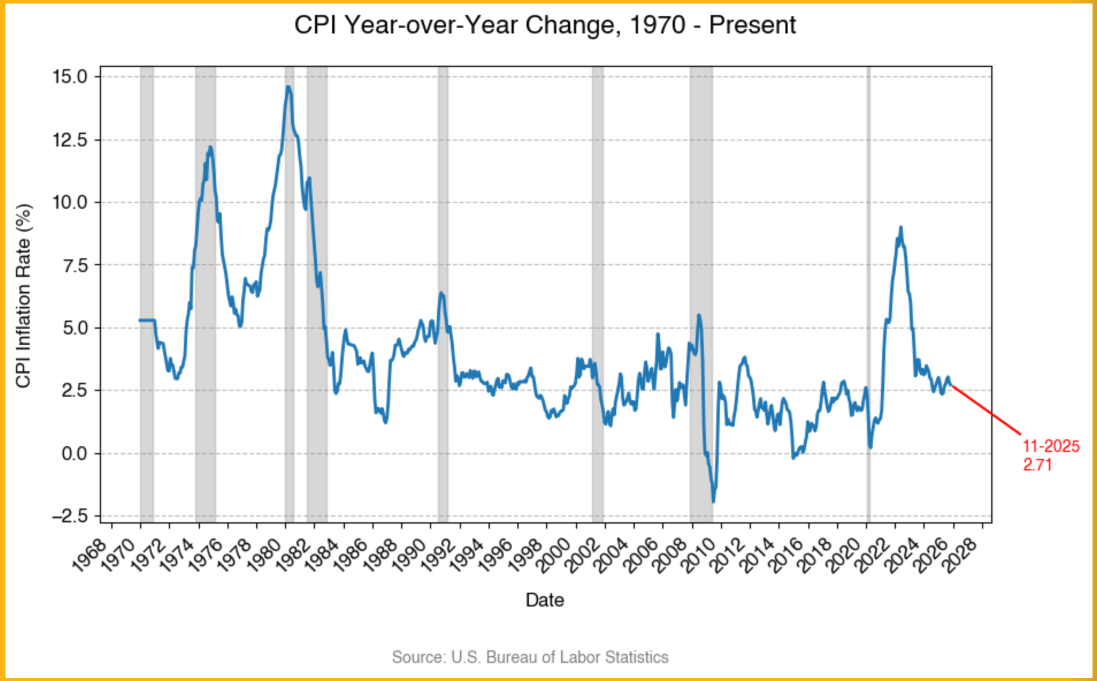
Unemployment Rate, 2020 - Present
Unemployment Rate (%)
14
12
10
8
6
4
11-2025
4.60
Oct 2019
Jan 2020
Apr 2020
Jul 2020
Oct 2020
Jan 2021
Apr 2021
Jul 2021
Oct 2021
Jan 2022
Apr 2022
Jul 2022
Oct 2022
Jan 2023
Apr 2023
Jul 2023
Oct 2023
Jan 2024
Apr 2024
Jul 2024
Oct 2024
Jan 2025
Apr 2025
Jul 2025
Oct 2025
Jan 2026
Date
Source: U.S. Bureau of Labor Statistics


Initial Unemployment Claims, 1970 - Present
Initial Claims
6000000
5000000
4000000
3000000
2000000
1000000
0
1968
1970
1972
1974
1976
1978
1980
1982
1984
1986
1988
1990
1992
1994
1996
1998
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
2026
2028
Date
12-2025
214000.00
Source: U.S. Employment and Training Administration

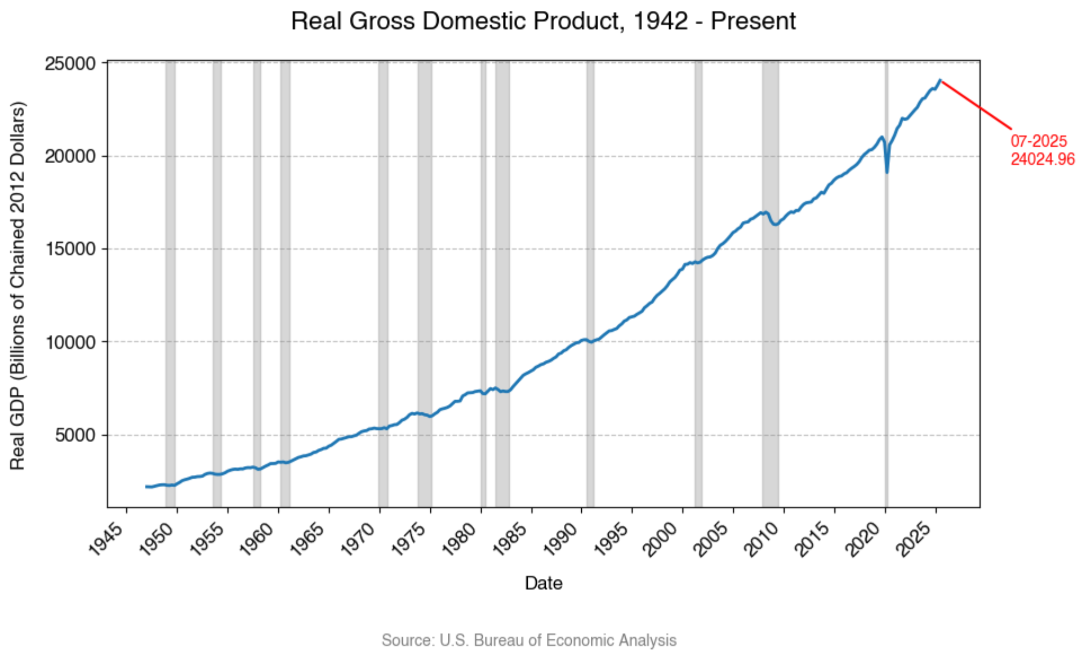
Job Openings per Unemployed Worker, 2001 - Present
Job Openings per Unemployed Worker
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
2026
Date
10-2025
1.01
Source: U.S. Bureau of Labor Statistics

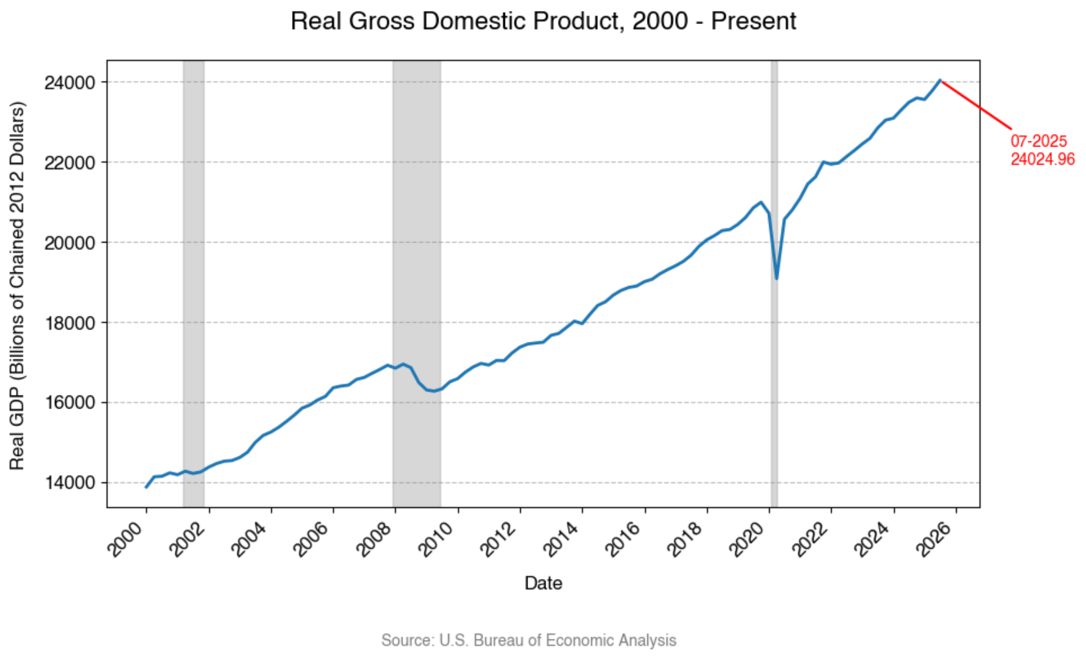
CPI/Inflation, 1970 – Present





Gross Domestic Product



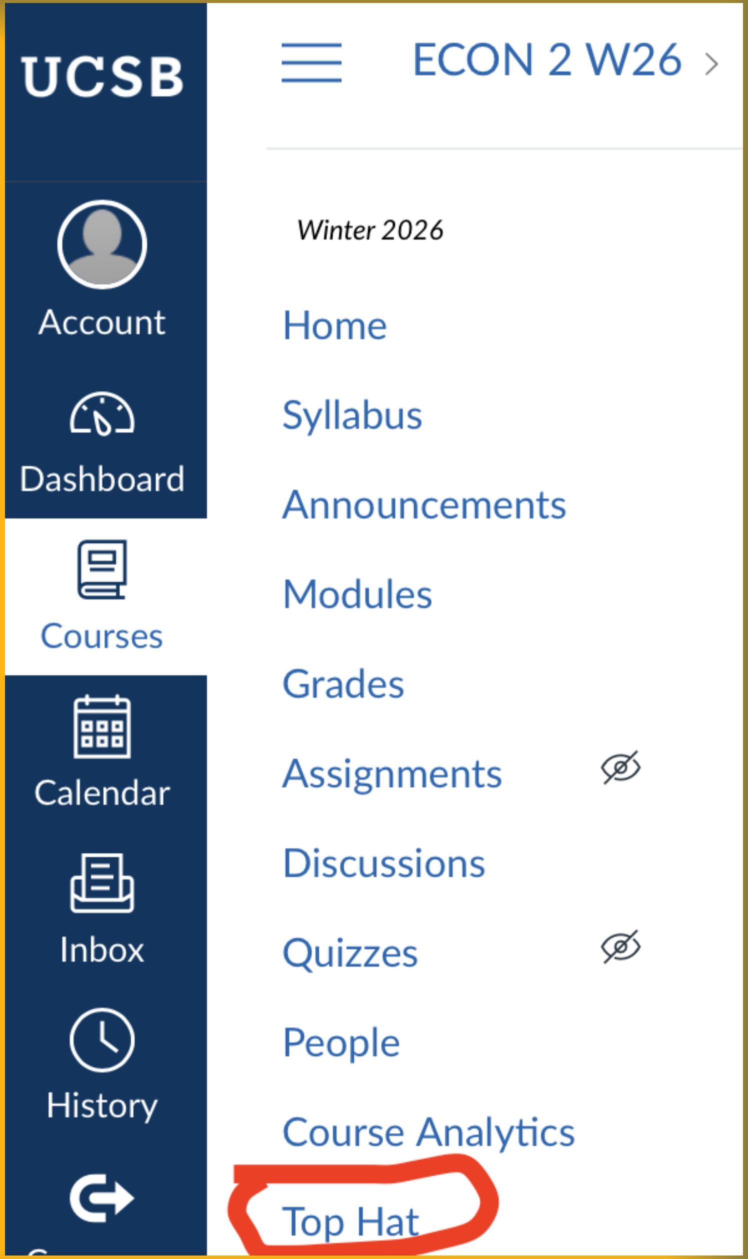


Course Materials

- **Textbook with TopHat subscription**
  - **Lower fee of $25 (contact me if you have financial constraints)**
  - **In-Class Questions**
- **Textbook Written by…me!**
  - **In the process of updating the book**
  - **Need your feedback!**
- **Traditional Textbooks…dated, but: Macroeconomics: Principles & Applications by Hall & Lieberman (cheap on Amazon, used)**

TopHat Interactive Classroom

- Instructions:
- Go to the Canvas Page for Econ 2
- Click on the TopHat link
- Create an account if needed
- If you click "Next" enough, the price should only be **$25 for a 4-month subscription.**



Course Grading

- Recent Confusion about Grading in Pre-Major Courses
- Department Grading Guidelines Instituted this Year
  - ~20% of students with A, A- (~140 students)
  - ~35% of students with B's (~245 more students)

Economics Department Grading Consistency Statement
In this course, grades will be assigned based on the following typical distribution:
A 20%, B 35%, C+ 10%, C 20%, C- or less 15%

- Course difficulty is set up to achieve this distribution
- I will not "curve" down, but may curve up and the minimum grade you can receive will still follow a standard grading scale

A: 93 – 100%	B+: 88 - 89%	C+: 78 - 79%	D+: 68 - 69%	F: <60%
A-: 90 - 93%	B : 83 – 89%	C: 73 - 78%	D: 63 – 68%	
	B-: 80 – 83%	C-: 70 – 73%	D-: 60 - 63%	

Course Grading

Baseline Portion (30% of total grade)

Weekly Lecture Quizzes (10%)

- Starting in week 1, released Wednesdays, due before next lecture (most Mondays)
- Participate in lecture (>50% of in-class questions answered)
  - Each lecture quiz question is worth 0.25% of your total grade, up to 10% (40 questions)
- Do not participate (<50% of in-class questions answered)
  - Each lecture quiz question is worth 0.2% of your total grade, up to 10% (50 questions)

Discussion Activities (15%)

- Weeks 2, 3, 6, 8, 10
- Top 3 of 5 activities worth 5% each, 15% total
- Gamification/AI in economics

Participation (5%)

- Don't be a story

Midterm Exam (30% of total grade)

Midterm Exam, April 27th

- 30 Multiple Choice Questions
  - ~20 - 23 Questions align relatively closely to Practice Questions, Exams, Quizzes, etc.
  - ~7 – 10 Questions use tools you have learned about on questions that are new/different
  - Ranging from small extensions to difficult applications
- If you miss the midterm with an excused absence, your final may be worth 70% of your total grade

Final Exam (40% of total grade)

Final Exam, June 9th

- 40 Multiple Choice Questions
  - ~28 to 30 Questions align closely with Practice Questions, Exams, Quizzes, etc.
  - ~10 to 12 Questions use tools you have learned about on questions that are new/different
  - Ranging from small extensions to difficult applications
- If you miss the final exam with an excused absence, you may receive an incomplete for the course and take the final in the fall quarter

Grading Thoughts and Policies

- GPA Requirement for Pre-major courses ←→ Athletic Race
  - Limited spots at the top, not everyone can get an A or B regardless of effort
  - Participation does not guarantee success
  - There are no “do-overs” or extra credit opportunities (30% of the course is “free”)
    - Special assignments are unfair to your peers and creates inequities, please don’t ask
    - Professional environments do not offer make-up work after deadlines
  - Doing the majority of training the night before will lead to a poor performance
    - Coming in stressed in week 9 or later is too late

- Goal is to
  - Reward learning – if you make learning a much as possible your goal, you will do well
  - Prepare you for success in upper division economics courses

Other Thoughts

- **“What if I have an emergency?!”**
  - First, reassess if it's an emergency: a breakup is not equivalent to a relative being diagnosed with cancer.
  - If it is an emergency, prioritize class appropriately. My experience is that students with the most serious emergencies do not care about their course points, they want support. I am happy to always offer that support!
  - Lots of “drops”…you can still get full credit on quizzes and activities if you start doing them by week 5!
  - If you notify me before the midterm with a ***documented reason for missing*** → the final exam may be worth 70% of your grade
  - If you notify me before the final with a ***documented reason for missing*** → you will get an incomplete in the course and take the final in the fall quarter

- **Do not let grades impact your self-worth or well-being**
  - Everyone is used to being at the top of their class
  - Employers are becoming frustrated with grade inflation, the lack of problem solving and work ethic of new hires.
  - Interview = Work Assignments = Grades are not an indicator of success
  - I don't know what your plans are for the future, but the likelihood of anyone's grade in their Intro to Macro course making a difference in their outcome in life is nearly 0%. Not figuring out how to prioritize learning can be costly.
  - “If I don't get my C+ bumped up to a B+, I won't get into the major.” That may be the best thing that happens!

Important Dates

- Lecture Quizzes done on TopHat
  - Due before the first lecture of weeks 2, 3, 4, 5, 7, 8, 9, and 10
  - Last Quiz due before the Final Exam on 3/18
  - Get 40 to 50 questions correct for full credit (10%)

- Discussion Section Activities
  - AI/Gamification Activities in weeks 2, 3, 6, 8, 10
  - Google Form Quizzes (simulate difficult exam questions) in weeks 1, 4, 5, 7, 9
  - Top 3 of 5 Activities needed for full credit (15%)

- Midterm on Monday, April 27th (Week 5)
  - 30 Multiple Choice Questions (30%)

- Final Exam on Wednesday, June 9th (12 – 2pm)
  - 40 Multiple Choice Questions (40%)

Federal Reserve is scheduled to meet on:
April 28th/29th (week of midterm)
June 16th/17th (one week after final)

Discussion Section Activity Example

- Week 2 (Activity #1): Housing Market Analysis
  - In groups of 2-3, choose a state, navigate to the Redfin Data Center
    - https://www.redfin.com/news/data-center/
  - Under Monthly Housing Market Data, download chosen state-level data
  - Upload data to Discussion Activities Website:
    https://matthewdlang18.github.io/principles-of-macroeconomics/activities/activity1/index.html
  - Examine 4 housing metrics (price, months of supply, days on market, price drops, sale-to-list)
    - Provide weights for each metric
    - Evaluate if the metric is signaling a balanced, buyers, or sellers market
    - Calculate a value for the health of the housing market
  - Feed data it into AI platform and ask for analysis
  - Compare your analysis to AI
  - Input results onto Google Sheet and TA will aggregate results on a separate webpage
  - Short reflection questions

- Discussion Activities will be posted on Website, Canvas and TopHat before sections

A Note on AI Usage

- Economics/Finance can handle AI better than other subjects
  - Lots of uncertainty in the subject
- AI is powerful if it is combined with critical thinking skills
- AI can create things quickly…if you know what to do
- But learning a new topic takes time and need to move slow

- Think of this course as having an AI Component and non-AI Component
  - Problem Sets, Exams = Critical Thinking Skills (non–AI)
  - Project= What can we accomplish with these new tools? (AI)

- If you find that using it is stopping you from thinking and learning, step away!

AI is Too Agreeable

- https://chatgpt.com/share/695583fc-7450-8012-9cea-a4c8961959e1

Possible AI Tools Used This Quarter

- Each week in Discussion Section there will be a small group activity merging course topics with an AI activity. You will turn in something by the end of each section.
  - ChatGPT: https://chatgpt.com
  - Perplexity: https://www.perplexity.ai
  - Claude: https://claude.ai/new
  - Google Gemini and Notebook LM (need personal account): https://gemini.google.com/app and https://notebooklm.google.com
  - Mistral AI: https://mistral.ai
  - Probably more! Let me know if you find good ones!
- Course Textbook is Connected to “Ace”
  - Probably not the best AI
- Top 3 of 5 activities count towards grade

Additional Resources

- Each week, difficult short-answer questions on the material will be posted
- The questions will be not be covered in your discussion section and instead covered every: TBD
- Review sessions for the midterm and final will also take place at these times
- Textbook! Textbook! Textbook!
- Make up your own extensions of questions!
- Administrative questions should be directed to the professor or Head TA
- Content questions should be asked in-person (lots of office hours)

Microeconomic Markets that Represent the Macroeconomy

- Technology
- Alcohol
- Gasoline
- Food
- **<u>Housing</u>**

How does the Housing Market Work?

- You budget $2000 per month for housing. What is the most expensive home/condo you can purchase?

Component	6% Interest Rate		
Home Price	$417,000		
20% Down Payment	$83,400		
Mortgage (Amount Borrowed/Principal)	$333,600		
Monthly Payment for 30 years	$2,000		
Mortgage Payments (360 Months)	$720,000		
Interest Paid after 360 Months	$386,400		
Mortgage + Down Payment	$803,400		

How does the Housing Market Work?

- You budget $2000 per month for housing. What is the most expensive home/condo you can purchase?

Component	6% Interest Rate	3% Interest Rate	
Home Price	$417,000	$417,000	
20% Down Payment	$83,400	$83,400	
Mortgage (Amount Borrowed/Principal)	$333,600	$333,600	
Monthly Payment for 30 years	$2,000	$1,406	
Mortgage Payments (360 Months)	$720,000	$506,160	
Interest Paid after 360 Months	$386,400	$172,560	
Mortgage + Down Payment	$803,400	$589,560	

How does the Housing Market Work?

- You budget $2000 per month for housing. What is the most expensive home/condo you can purchase?

Component	6% Interest Rate	3% Interest Rate	3% Interest Rate
Home Price	$417,000	$417,000	$593,000
20% Down Payment	$83,400	$83,400	$118,600
Mortgage (Amount Borrowed/Principal)	$333,600	$333,600	$474,400
Monthly Payment for 30 years	$2,000	$1,406	$2,000
Mortgage Payments (360 Months)	$720,000	$506,160	$720,000
Interest Paid after 360 Months	$386,400	$172,560	$245,600
Mortgage + Down Payment	$803,400	$589,560	$838,600

How does the Housing Market Work?

Component	6% Interest Rate	3% Interest Rate	3% Interest Rate
Home Price	$417,000	$417,000	$593,000
20% Down Payment	$83,400	$83,400	$118,600
Mortgage (Amount Borrowed/Principal)	$333,600	$333,600	$474,400
Monthly Payment for 30 years	$2,000	$1,406	$2,000
Mortgage Payments (360 Months)	$720,000	$506,160	$720,000
Interest Paid after 360 Months	$386,400	$172,560	$245,600
Mortgage + Down Payment	$803,400	$589,560	$838,600

- Is this ever a good deal? What are we missing?

How does the Housing Market Work?

30-Year Mortgage vs. Rental Apartment at $2,000/month

Component	6% Interest Rate	Rental Apartment
Home Price	$417,000	$0
20% Down Payment	$83,400	$0
Mortgage (Amount Borrowed/Principal)	$333,600	$0
Monthly Payment for 5 years	$2,000	$2,000
Total Monthly Payments (60 Months)	$120,000	$120,000
Monthly Payments + Down Payment	$203,400	$120,000
Principle Owed after 5 years	$310,429	$0
Sale Price	$450,000	$0
Savings after Sale (Renting = 4% US Bond)	$139,571	$101,468

How does the Housing Market Work?

- The home is a place to live AND a place to store your money (investment)
- What are we missing?

Component	6% Interest Rate	Rental Apartment	3% Interest Rate
Wealth after 5 Years	$139,571	$101,468	$153,409
Homeowner Taxes ($4,500/year)	$22,500	$0	$22,500
Homeowner's Insurance ($2,500/year)	$12,500	$0	$12,500
Upkeep ($100/month)	$6,000	$0	$6,000
Realtor Fees (up to 5% of sale price)	$22,500	$0	$22,500
Sale Price of $440 instead of $450K	$10,000	$0	$10,000
Extra Costs	**$73,500**	**$0**	**$73,500**
Invest Monthly Savings into 4% bond			$39,911
Savings Account after Adjustments	$66,071	$101,468	$119,820

How does the Housing Market Work?

- **A lower interest rate makes buying more affordable!**
  - **Decreasing the interest rate can incentivize a homebuyer to purchase a home**
- **Home values are important to household wealth**
  - **Important later in the term**
  - **Rising home values increase homeowner wealth**
- **Higher home prices make renting more desirable for potential home buyers**
- **Lower home prices make renting less desirable for potential home buyers**

Demand for Housing

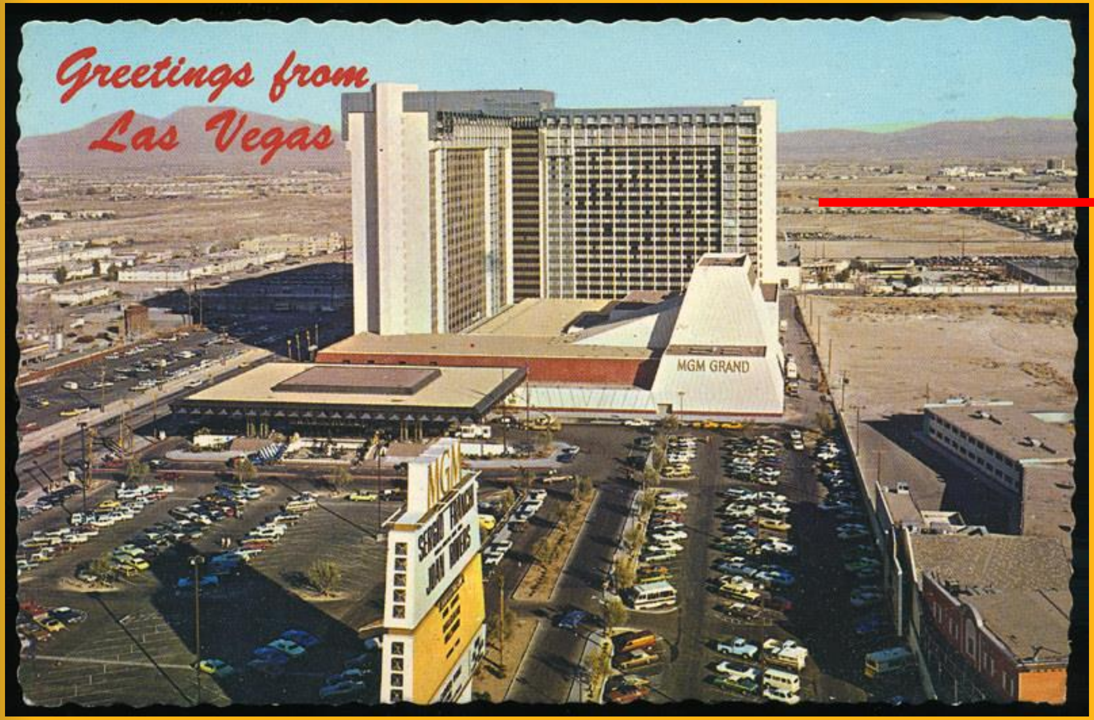
Supply of Housing

- How is housing supply determined?

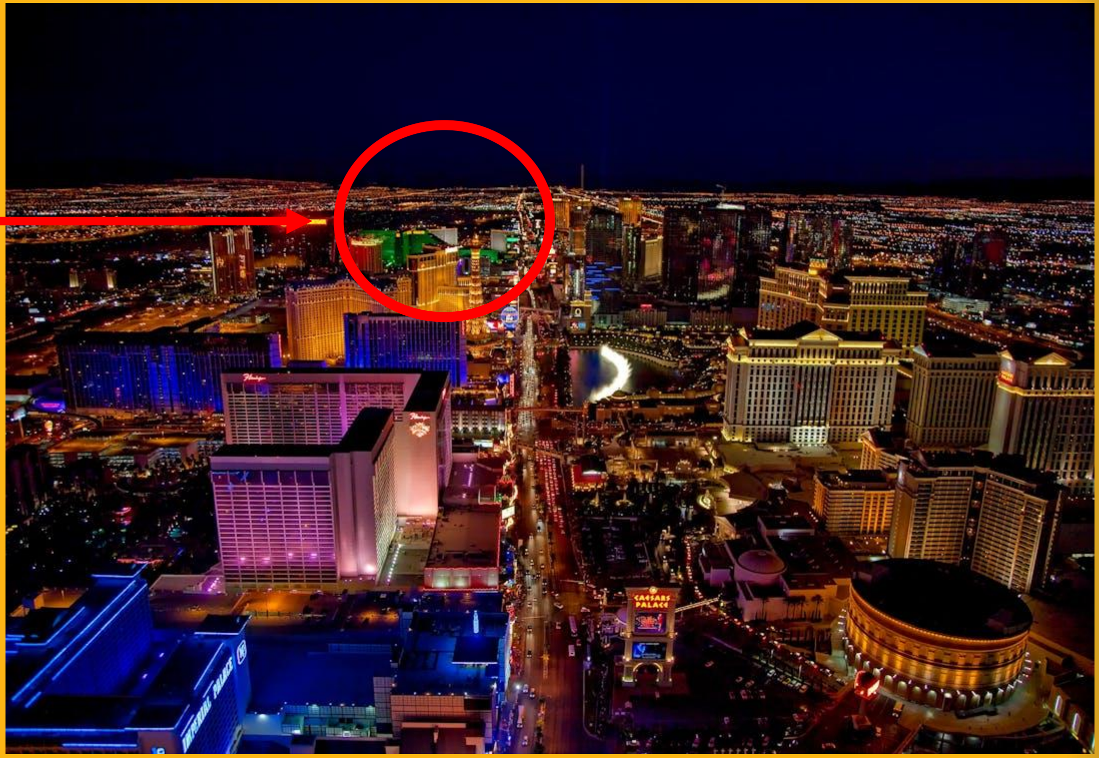
Housing Market Equilibrium

Explaining Changes in the Housing Market

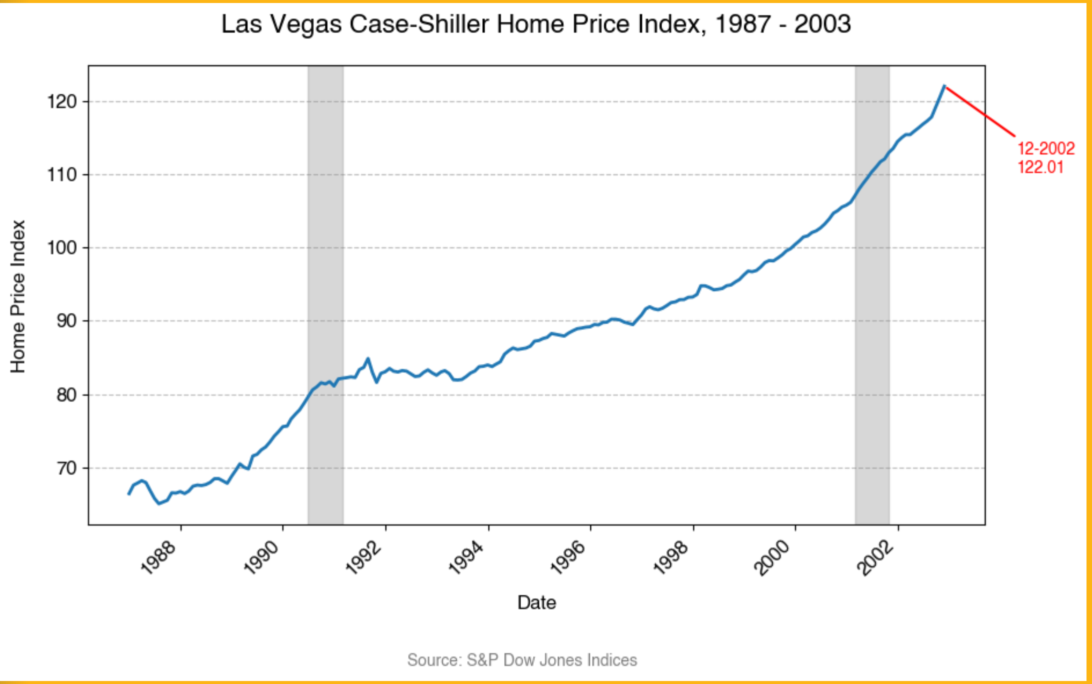
MGM Grand, Las Vegas, 1973



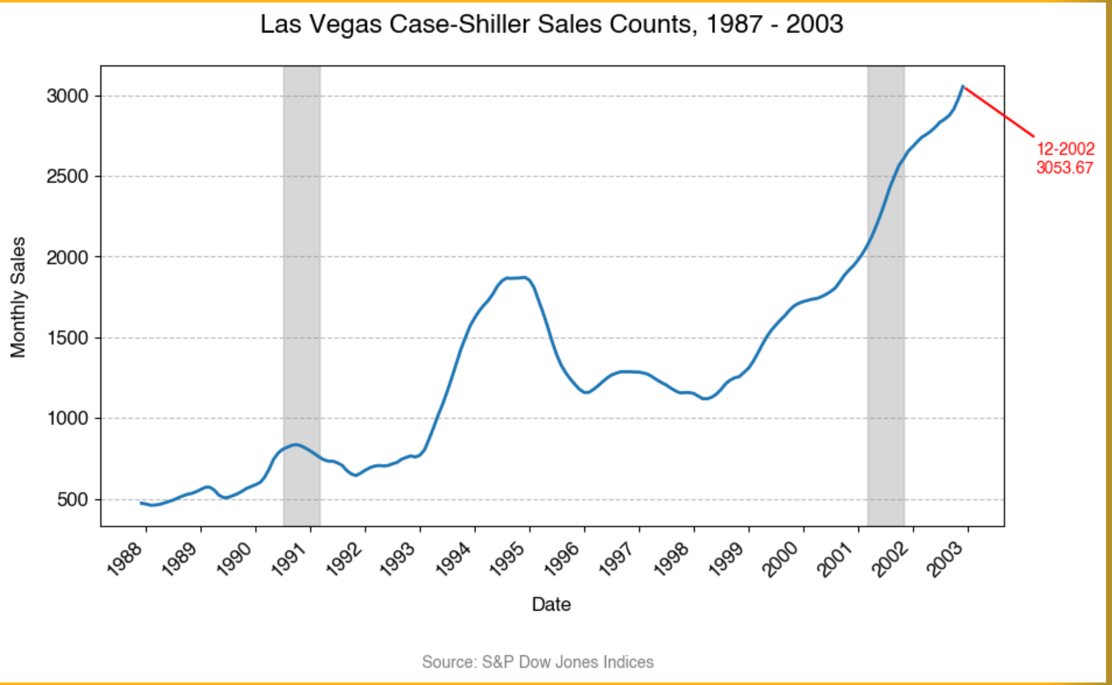
Las Vegas Strip, 2024



Explaining Changes in the Housing Market



Las Vegas housing prices ~ doubled between 1987 and 2003



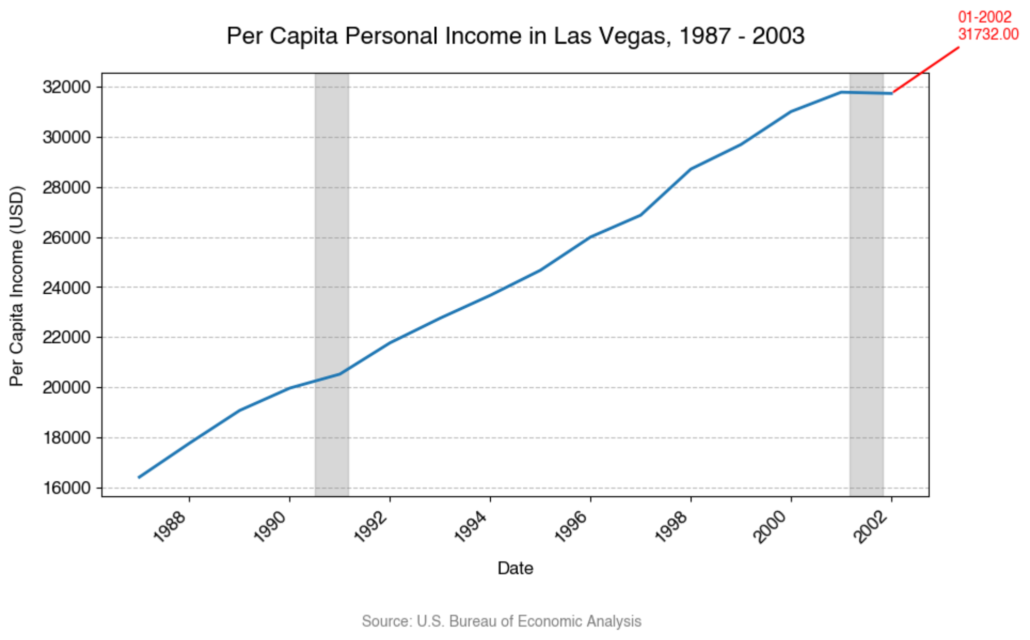
Las Vegas homes bought/sold increased from 500 per month to 3000 per month between 1987 and 2003

Housing Market Equilibrium

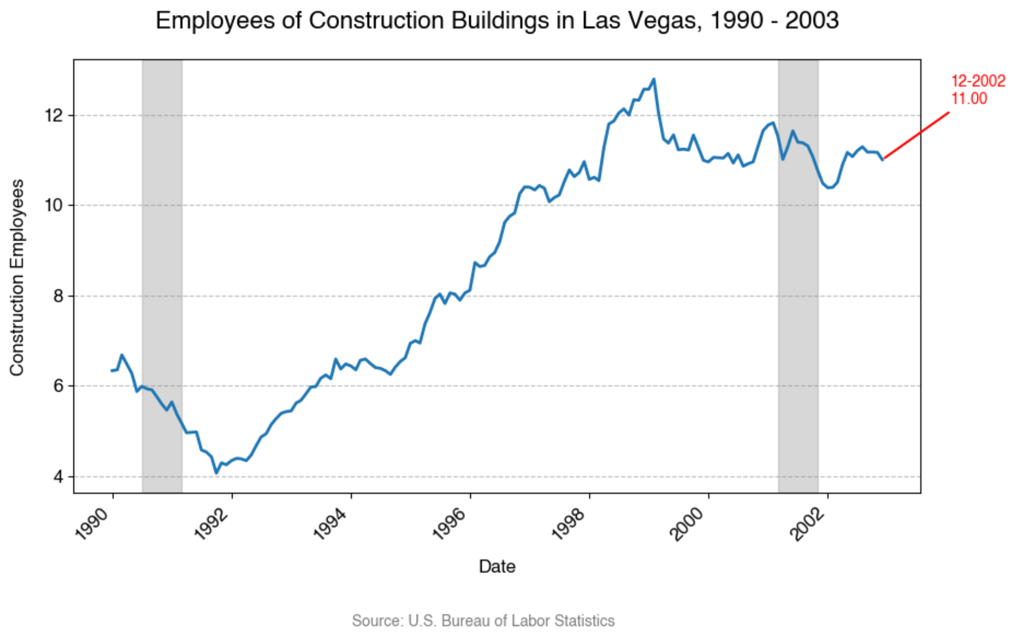
Q: Between 1987 and 2003, prices and quantity of homes increased. What is the most likely reason for this?

Unemployment Rate, Las Vegas, 1990 - 2003
Unemployment Rate (%)
Date
12-2002
5.80
Source: U.S. Bureau of Labor Statistics

30-Year Fixed Mortgage Rate, 1987 - 2003
Mortgage Rate (%)
Date
12-2002
5.93
Source: Freddie Mac



Per Capita Personal Income in Las Vegas, 1987 - 2003
Per Capita Income (USD)
Date
01-2002
31732.00
Source: U.S. Bureau of Economic Analysis



Employees of Construction Buildings in Las Vegas, 1990 - 2003
Construction Employees
Date
12-2002
11.00
Source: U.S. Bureau of Labor Statistics

What happened between 2003 and 2007?

Mortgage-Backed Securities

Mortgage 1: 100K

Mortgage 2: 200K

Mortgage 3: 300K

1 million over 30 years

750K

What happened between 2003 and 2007?

Mortgage-Backed Securities

Mortgage 1: 100K

Mortgage 2: 200K

Mortgage 3: 300K

1 million over 30 years

750K

What happened between 2003 and 2007?

- Banks could sell MBS to investors→ more funds to lend
- Constraints to buying a home?
  - **Down-payment**
  - Interest rate
  - Income requirements

CNN Money

YOUR MONEY > Your Home

SAVE THIS | EMAIL THIS | PRINT THIS | MOST POPULAR

Home buying with no money down

If the only thing standing between you and homeownership is a downpayment, consider your options.

December 23, 2003: 5:28 PM EST

By Sarah Max, CNN/Money Staff Writer

BEND, Ore. (CNN/Money) � Up until a year ago, photographer Melissa Jansson, 30, planned to rent for the foreseeable future. At that point, the few hundred dollars she had in savings wouldn't even cover her closing costs.

In November, Jansson closed on a $150,000 home of her own with a 3 percent downpayment she scored through a grant from the state of Oregon for qualified first-time homebuyers. Even better, she found a seller willing to pay all but $12 of her closing costs.

"Until I heard about the grant program buying a house was a fantasy," said Jansson, who shares her two-bedroom, two-bathroom home with a roommate and ultimately pays only $40 a month more to own than she did to rent.

Very little down

"No-money-down home purchases used to be the kind of thing you only saw on late night TV," said Keith Gumbinger, vice president for HSH Associates.

Now they're in the mainstream.

What happened between 2003 and 2007?

- Banks could sell MBS to investors→ more funds to lend
- Constraints to buying a home?
  - Down-payment
  - **Interest rate**
  - Income requirements

THE WALL STREET JOURNAL.

Latest World Business U.S. Politics Economy Tech Markets & Finance Opinion Arts Lifestyle Real Est

Teaser Rates On Mortgages Approach 0%

As Lending Frenzy Slows, Banks Add Twists to Lure Homeowners; Many of the Deals Have Pitfalls

By Ruth SimonStaff Reporter of THE WALL STREET JOURNAL

Feb. 15, 2005 at 12:01 am ET

Share Resize

Taking a page from credit-card companies and car makers, mortgage lenders are touting loans with rock-bottom introductory rates -- in one case, nearly 0%.

Most of these loans are so-called option adjustable-rate mortgages, which carry an initial rate as low as 1%. One key and unusual feature: Borrowers get up to

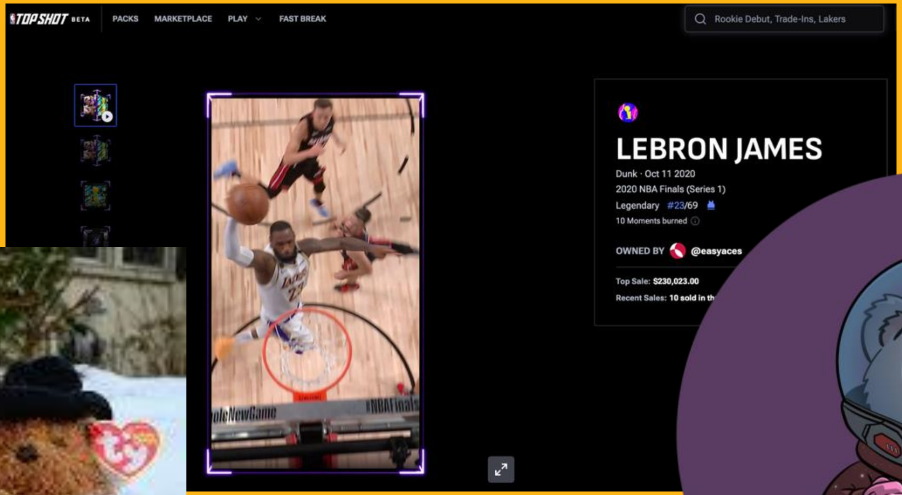
What happened between 2003 and 2007?

- Banks could sell MBS to investors→ more funds to lend
- Constraints to buying a home?
  - Down-payment
  - Interest rate
  - **Income requirements**
  - **Alt-A Loans**
    - **No documents**
    - **NINJA Loans (no income, no job or assets)**

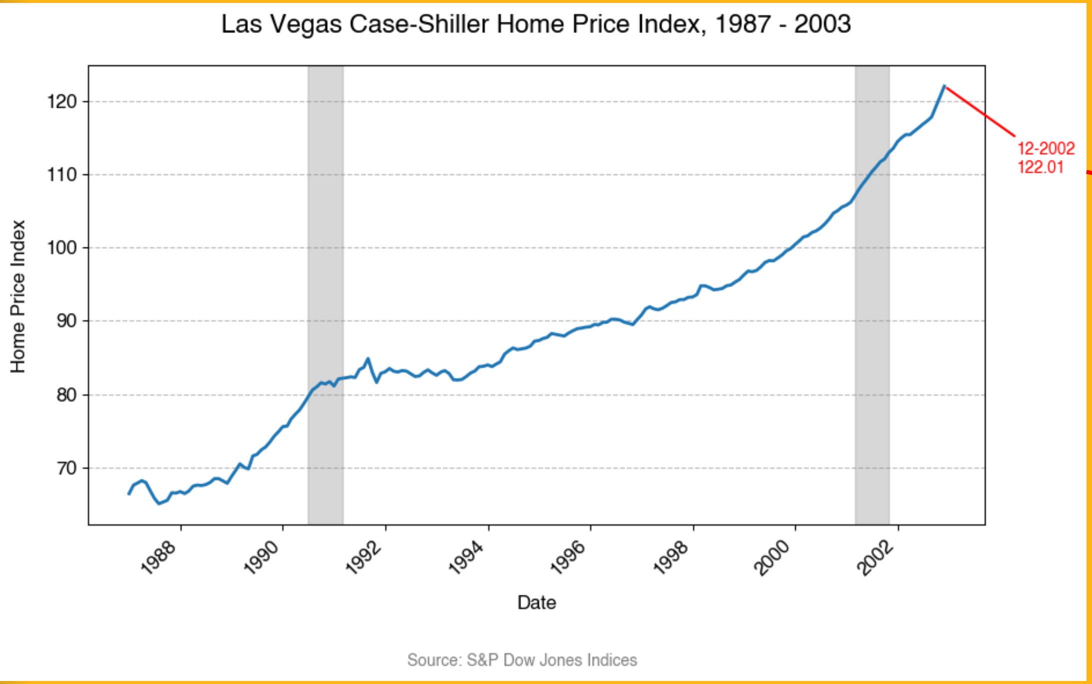
Year	**Share of Alt-A loans (in %)**
2001	2.8
2002	2.5
2003	2.3
2004	7.7
2005	13.8
2006	15.9

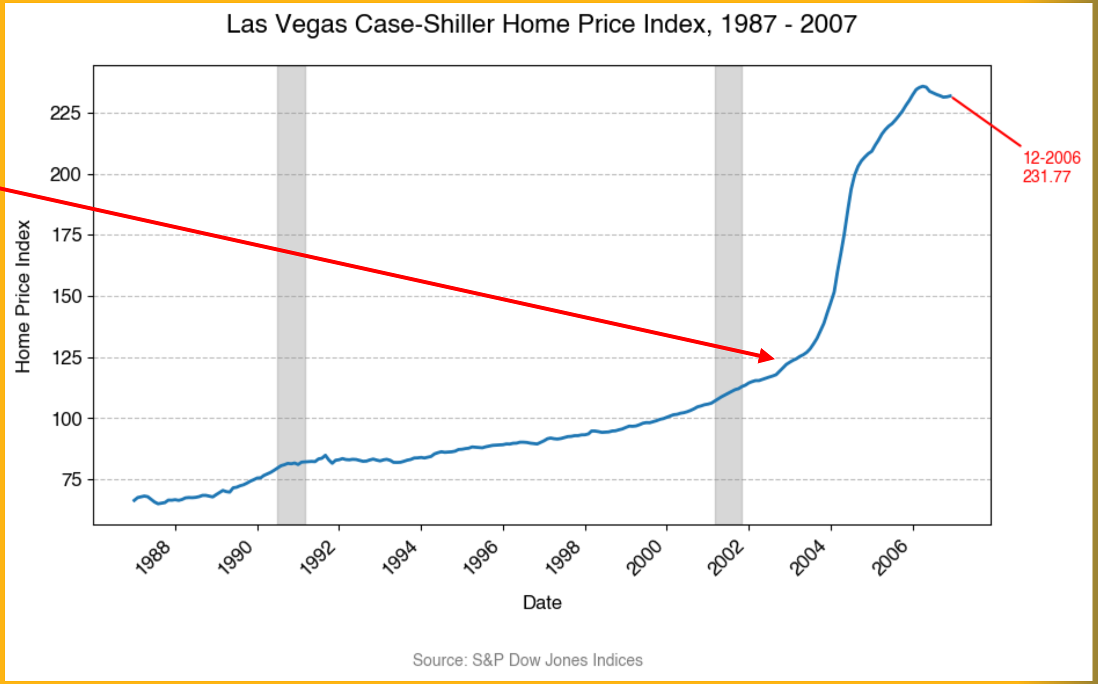
What happened between 2003 and 2007?

- No down-payment, low interest rate, limited income requirement?
- Buy a home, wait for the price to increase, sell the home, profit, repeat
- Buying an asset with the purpose of reselling (not using) = speculation



Explaining Changes in the Housing Market

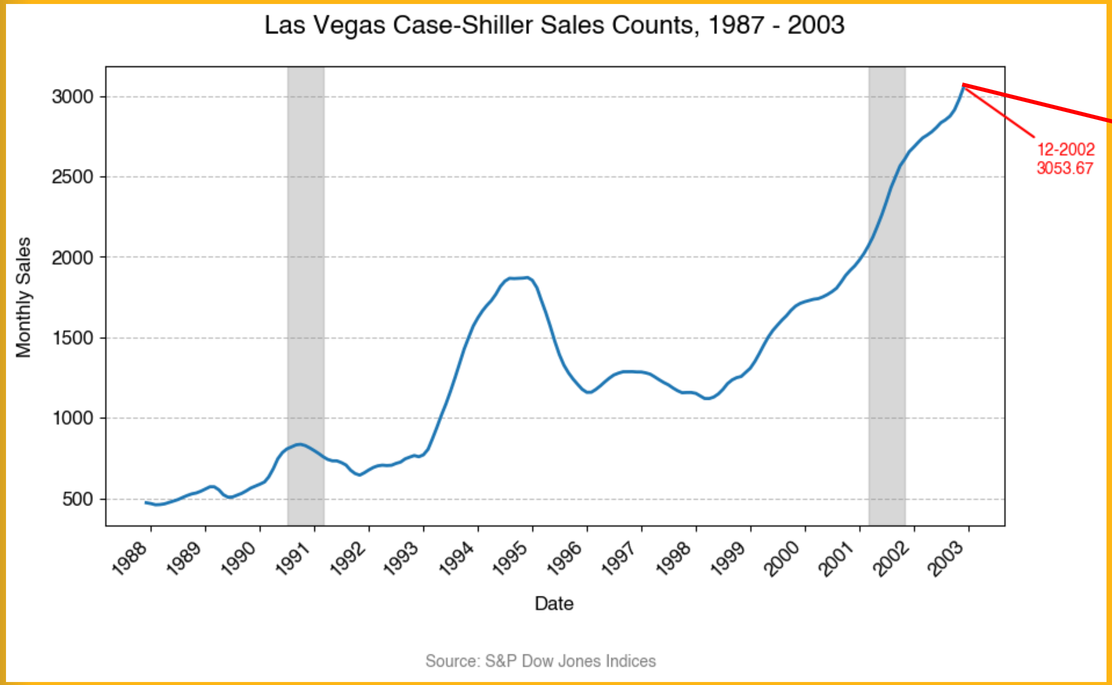


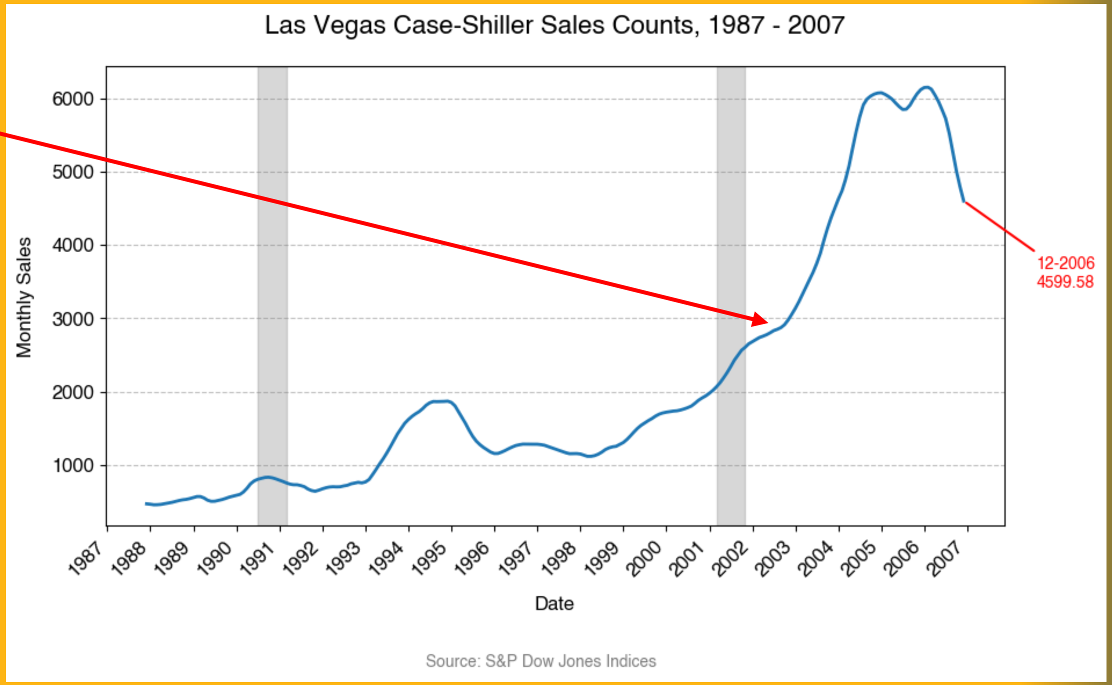


Las Vegas housing prices ~ doubled between 1987 and 2003 (16 years)

Las Vegas housing prices ~ doubled between 2003 and 2007 (4 years)

Explaining Changes in the Housing Market





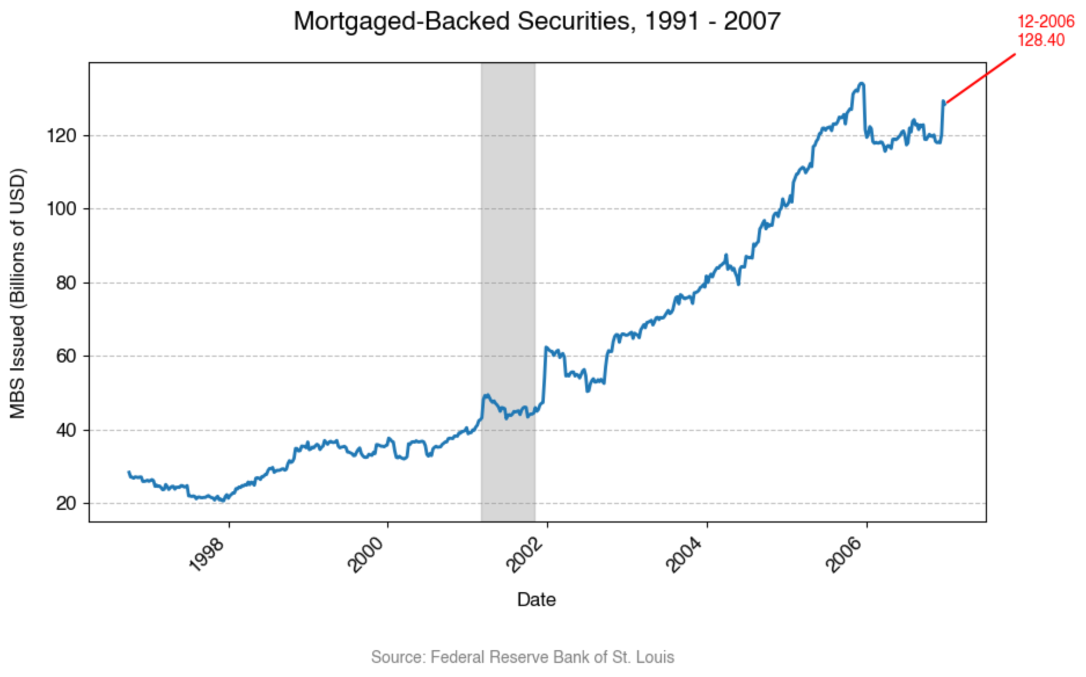
Las Vegas homes bought/sold increased from 500 per month to 3500 per month between 1987 and 2003

Las Vegas homes bought/sold increased from 3500 per month to 7000 per month between 2003 and 2005/2006

Decreased back to 4600 per month by 2007

What happened between 2003 and 2007?

- Investors piled into MBS
- MBS split into different levels of risk (Collateralized Debt Obligations, CDO)
- Financial firms insured MBS with Credit Default Swaps (CDS = insurance if MBS fails)
  - Possible for pension funds and “safe” investors to join
  - Did not need to own MBS to purchase CDS
- By 2006, price growth muted, sales falling
- Adjustable-Rate Mortgage (ARM) teaser rate ends



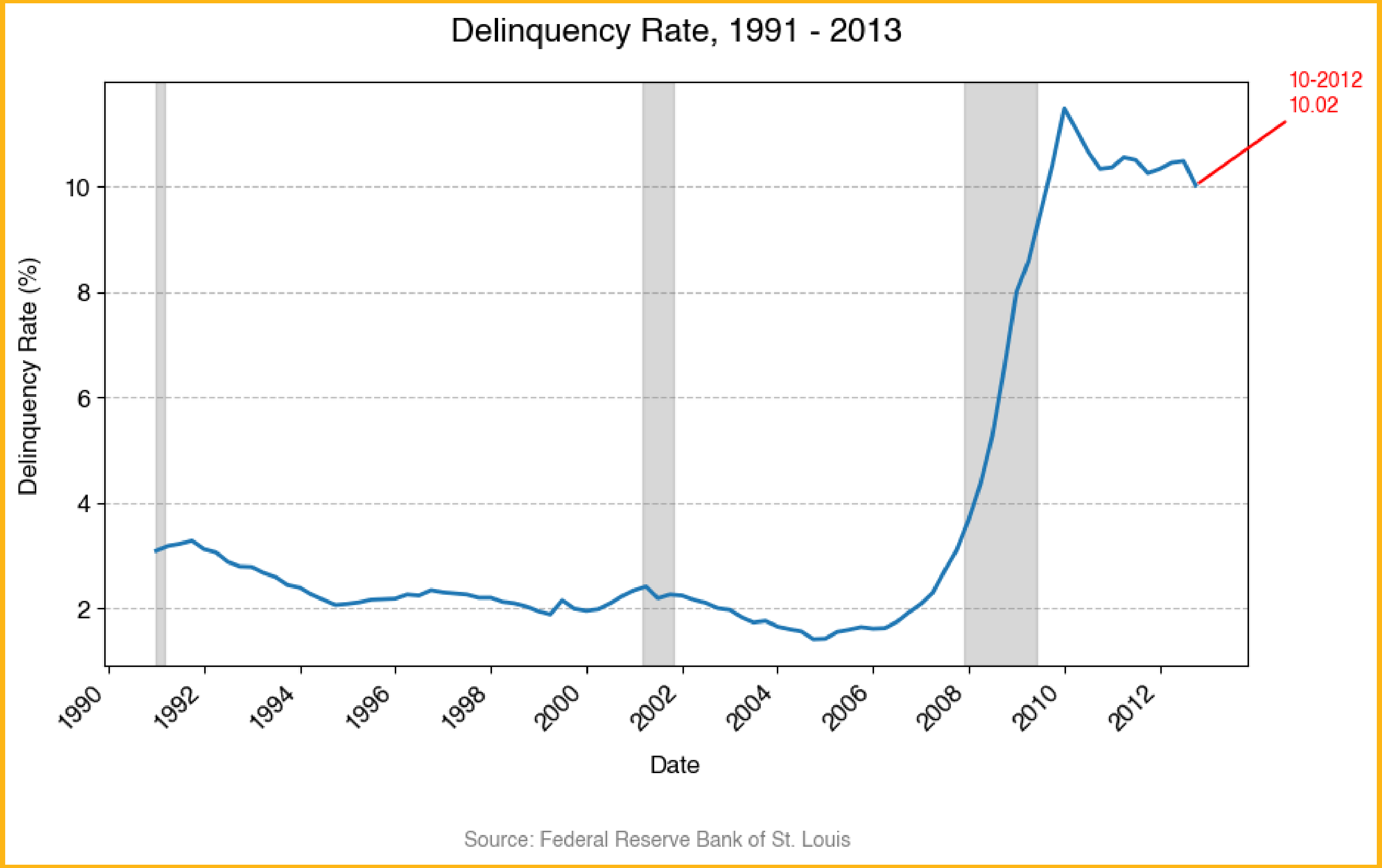
What happened between 2003 and 2007?

- Speculators?
- Those with multiple homes?
- Those thinking about buying?

What happened after 2007?

- Stagnant home prices + rising ARM
- Many homes owned by those with:
  - Low down-payment/no equity in home
  - Current home price below principal of the loan
  - Cost of holding home increasing
- Unemployment rising

What happened after 2007?



What happened after 2007?

- Mortgage-Backed Securities

Mortgage 1: 100K

Mortgage 2: 200K

Mo

00K

1 million over 30 years

750K

What happened after 2007?